

PRODUCT GUIDE

CFM Coach

A measurement and proposal system that sits beside a live forecasting agent. It records every run, waits for the market to resolve each forecast, scores the result against a random walk, and proposes calibration changes that a person accepts or rejects. Written for teammates who will run it, read its reports, and decide on its proposals.

Field	Summary
What it coaches	cfm_agent_v_5_2, a WTI crude forecaster that publishes P10 to P90 quantiles at 5, 10 and 21 business days.
What it changes	Numeric constants (interval width, overlay gain, random-walk anchor, engine settings, ensemble weights) and, through a challenger package, prompt text.
What it never does	Edit the agent package, apply a change without a human decision, or fit on anything except live, forward-dated runs.
Where it stands	288 recorded runs across three streams, 199 tests passing, one budgeted review run completed (\$0.89). No calibration has been accepted yet.

1 | THE PRODUCT

What the coach is for

The agent it coaches turns web research and an LLM's categorical judgment into a price distribution. Before the coach existed there was no way to know whether that distribution was any good, and no way to adjust the constants that produce it without editing a package whose provenance is pinned by a manifest. The coach solves both problems from the outside.

It has two loops. The daily loop is arithmetic only: it records each run, resolves horizons as WTI closes arrive, and scores four variants side by side. The weekly loop uses one LLM, on a hard budget, to read the resolved cases, tag repeated errors with pointers into the evidence, and draft ranked proposals. A proposal that a person accepts changes a coach-owned ledger and produces a coached forecast beside the agent's own. The agent's published forecast does not change.

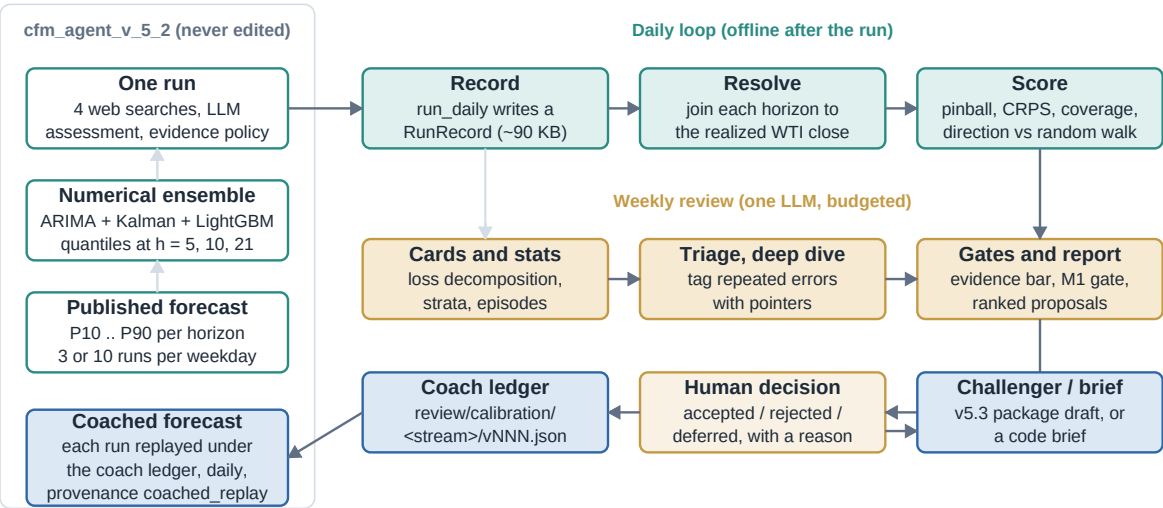


Figure 1. The agent on the left is read, never written. The daily loop runs offline after each run. The weekly loop and every human decision produce outputs on the right, which feed the coached forecast rather than the live agent.

The goal, in one line

Agent pinball loss below random-walk pinball loss at each horizon on live_forward rows, with 80% intervals that cover 80% of the time. Today the agent loses to the random walk by 14 to 18%, and most of that gap is in the numerical ensemble rather than in the LLM's judgment (section 6).

2 | DESIGN

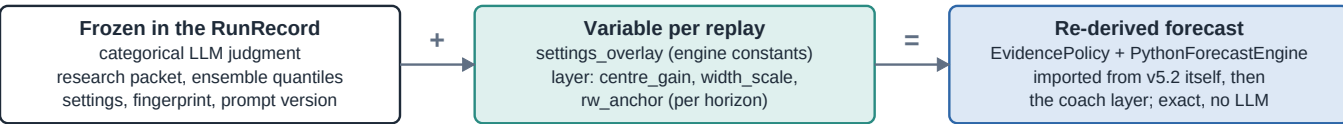
Five invariants and the replay that rests on them

Every design choice below follows from one fact: the coach must be able to answer "what would this forecast have been under different constants?" exactly, for every stored run, without spending a token. That is possible only because the LLM's output is categorical and stored, while the numbers are derived by code the coach can re-run.

Invariant	What it means in practice	Where it is enforced
The agent package is never edited	Calibration is injected as settings at construction time. Its MANIFEST.sha256 stays valid; a changed build changes the package fingerprint on every new record.	ledger.to_agent_settings; package_fingerprint on RunRecord
Only live_forward runs are evidence	A run re-executed at a past cutoff searches today's web, so its source selection is shaped by hindsight. Such runs are stored and shown but never fitted on.	Gate condition 1; CoachSettings.fitting_eligible_provenance
Nothing auto-applies	A passed gate verdict is a proposal. A person writes the decision into the proposal file; accept.py applies it to the coach ledger, never to the live one.	ComparisonVerdict is the only place passed=True can be built; accept.py refuses calibration_v52_*
One stream, one corpus, one ledger	A stream is one (agent package, LLM, runs per day). A different LLM is a different agent, so corpora and ledgers are kept apart on disk and the gate refuses a pooled fit.	streams.py; gate conditions 2 to 4
The LLM never touches a number	The agent's prompt has no numeric constants. It emits categories (large_up, substantially_wider) and Python owns every number. That is what makes replay exact and free.	v5.2's EvidencePolicy and PythonForecastEngine, imported by replay.py

Record and replay

A run record is a schema-versioned snapshot of one agent execution: the categorical assessment, the research packet, the per-model and ensemble quantiles, the settings in force, the package fingerprint, the prompt version, and the provenance. Replaying it means re-running v5.2's own policy and engine over the stored judgment under a chosen calibration, then applying the coach's layer. Those two classes are imported from the agent, not reimplemented, so the arithmetic cannot drift.



Fidelity check, run before any scoring:
replaying every record under its own calibration must reproduce the published quantiles bit for bit.
Today: all 288 records across three streams replay with max |error| = 0.0. Any drift halts the report.

Figure 2. A calibration version varies only the middle box. Because the left box is fixed, the right box is deterministic, and fidelity can be checked by replaying each record under its own calibration.

The post-engine calibration layer

The agent's own uncertainty multipliers fire only when the LLM asks for a width change, so a neutral run publishes the raw ensemble interval with no lever to adjust it. The layer gives the coach that lever without touching the package. It applies three per-horizon transforms in order; an empty layer is the identity, which is why baseline v001 reproduces the agent bit for bit.

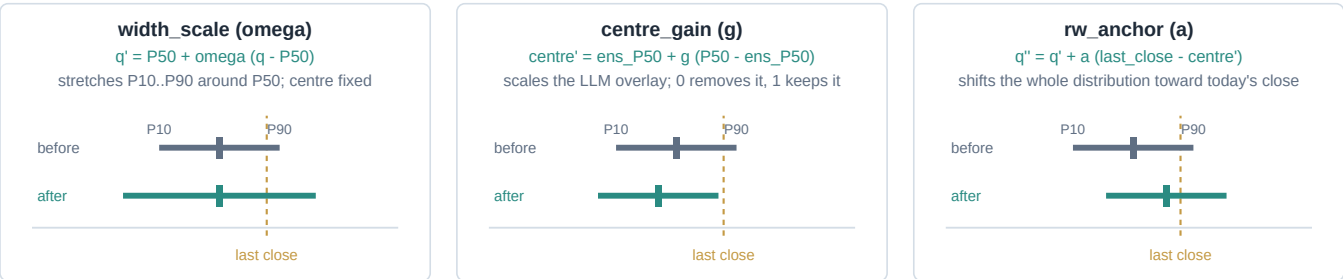


Figure 3. The three layer levers on one horizon. Grey is the published distribution, teal the replayed one. A non-finite or crossing quantile raises rather than publishing.

3 | WHAT GETS CALIBRATED

Seven levers, three delivery paths

"Calibration" here means any change the coach can express without editing the agent package. Five levers are numeric and land in a coach-owned ledger; two are text or code and leave the numeric path.

Lever	What it is	Fitted or argued on	Priced by	Delivered as
width_scale (omega)	Per-horizon multiplier on the P10..P90 span around P50. Attacks coverage: intervals cover about a third of outcomes against an 80% target.	Ensemble-only history (no LLM, no research, so no hindsight in source selection): the 80th percentile of the stretch each past forecast needed.	Exact replay	Coach ledger version
rw_anchor (a)	Fraction of the distance from the centre to the last close that the whole distribution is shifted. Attacks the ensemble's level bias.	Same history, by minimising pinball on the training window with a moving-block bootstrap for the interval.	Exact replay	Coach ledger version
centre_gain (g)	Scale on the LLM overlay, the distance the agent moved the centre away from the ensemble.	Live records only. Not fitted yet: same-cutoff runs show the overlay zeroed at random by citation bookkeeping, so a fitted mean would be biased toward zero.	Exact replay	Coach ledger version
settings_overlay	Any tunable constant of v5.2's evidence policy or engine, such as <code>large_action_width_fraction</code> . A denylist refuses identity fields: model names, timeouts, audit and code-execution flags.	Coarse grid, one constant per candidate, value argued in the proposal.	Exact replay	Coach ledger version
ensemble_weights	Simplex weights over ARIMA, Kalman and LightGBM. A random-walk member is dropped live, so that variant ships as a code brief.	Stored component quantiles.	Approximate recombination (LightGBM samples move with the seed)	Coach ledger version
Prompt, skill or persona text	Wording changes inside the agent, for example how claims cite sources.	Tagged cases with pointers into pre-cutoff material; two-unit evidence bar.	Not priceable offline	Challenger package v5.3_<slug>, run as its own stream, judged by pairing
Code or data	Changes only an engineer can make, such as fitting LightGBM on returns instead of levels.	Tagged cases plus the seeded backlog.	Sometimes, via <code>history_variant</code> re-runs	Written brief

4 | THE GATE

The only component allowed to say a change is better

Everything else in the coach measures; the comparison policy decides. It checks ten conditions in a fixed order and a candidate passes only if all ten hold. Eight are about what the corpus is and how much of it counts; the two statistical tests come last, because a p-value over a contaminated or pooled corpus launders the contamination into something that looks like evidence. One of the eight is `single_tunable`: a candidate may move one knob per cycle (`width_scale` at three horizons is one knob), because two moved together cannot be attributed.

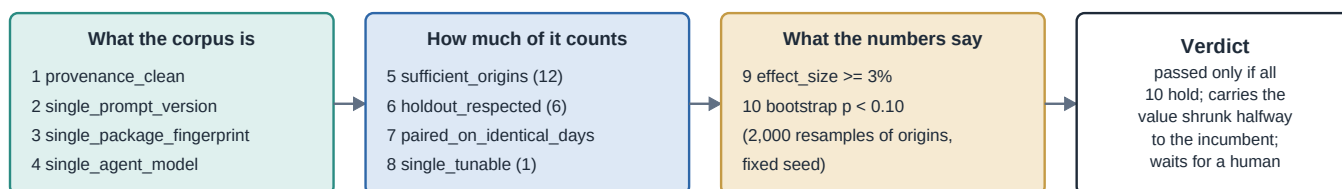


Figure 4. The gate as checked; thresholds in brackets are the CoachSettings defaults. Fidelity is a precondition run before it.

Sample size is counted in origins (cutoff dates), never in (origin, horizon) rows, because a run's three horizons share one assessment and overlapping outcome windows; the bootstrap resamples whole origins for the same reason. A passed verdict carries the candidate pulled halfway back toward the incumbent, because a point estimate from a small, autocorrelated sample over-corrects if applied in full.

5 | THE WEEKLY REVIEW

One LLM, eight stages, every count computed by code

The review agent exists because the scoreboard can say the agent is wrong but not why. It reads resolved cases, tags repeated errors, and writes proposals ranked by how much of the gap to the random walk they would close. It runs on gemini-3.5-flash through the team proxy, with per-stage reasoning effort, and a cost ledger that hard-stops the run at the weekly budget.

Stage	Who does the work	What it produces	LLM calls
A Collect	code	Records loaded, resolved and scored; which (stream, cutoff, horizon) keys are newly ready	0
B Cards	code	One case card per cutoff: representative run, dispersion across same-cutoff runs, numerical base table with the loss decomposition	0
C Stats	code	Strata, episodes, independent windows, priced counterfactuals; every count synthesis may quote	0
D Triage	LLM, structured output	Tags from the static taxonomy with pointers; code validates every pointer and drops what fails	1 per card
E Deep dive	LLM tool loop	For the two cards where the overlay is most implicated: verdict, hindsight driver, knowability; base-dominated cards skipped	K = 2 cards
F Synthesis	LLM tool loop	Hypothesis updates and proposal drafts; only keys returned by a counting tool may support a hypothesis	1 loop
G Gates	code	Evidence bar per track, rejected-fingerprint block, cross-stream contrast, M1 gate on numeric candidates	0 (critic optional)
H Render	code	report.md (source of truth), report.html, manifest with cost and probe result	0

Tracks and the evidence bar

Every tag carries a track, which decides what kind of proposal it can feed and what evidence it needs. Numeric and policy tags are judged by pricing on stored records plus the gate. LLM-judgment tags need pointers into pre-cutoff material for every supporting case, and enough support to rule out one lucky episode: outcome-dependent tags need three independent windows (cutoffs at least h business days apart) across two episodes of opposite sign; process tags need four distinct cutoffs plus the fraction of same-cutoff runs that show them. Hindsight text is stored separately and can never be rendered into a proposal.

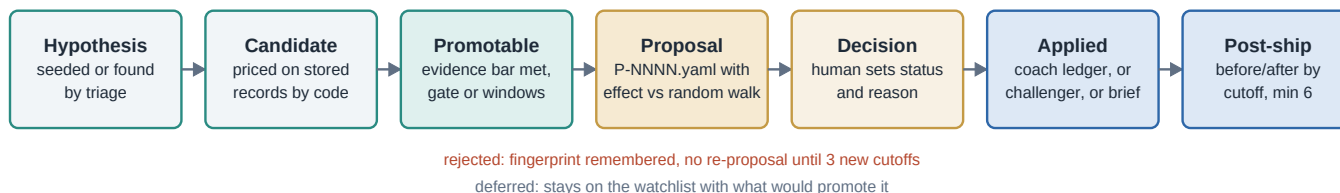


Figure 5. Life of a proposal. Only the decision block is edited by a person; every other state is written by code or by the weekly job.

Memory

The coach's memory is files, append-only where the LLM writes and human-edited only where a person decides: annotations per stream, a hypothesis event log, proposal YAMLS, a static taxonomy of 22 error codes, a seeds file that loads the IMPROVEMENTS backlog as unverified hypotheses (never shown to triage), and a playbook the coach reads but does not edit. A proposal rejected by the team is remembered by fingerprint and cannot be re-proposed until three new cutoffs exist.

6 | USING IT

Commands, cadence, and how a decision is made

Only the `run_daily` family and the real weekly review call an LLM. Everything else is arithmetic over records that already exist. Run from the repo root, because the price cache resolves relative to the working directory.

What you want	Command (from the repo root, prefixed <code>uv run python -m energy_oil_forecasting.cfm_coach.</code>)	LLM	Cost / time
The scoreboard	<code>report [--stream=v52_lite]</code>	no	free, 2 s
Check the checkout	<code>pytest implementations/energy_oil_forecasting/cfm_coach/tests -q</code>	no	free, 70 s
One live forecast, recorded	<code>run_daily [YYYY-MM-DD] [--stream=v52_lite]</code>	yes	5 calls, 4 min
What the 08:00 weekday job runs	<code>run_daily_all</code>	yes	13 runs, 50 min
Re-run v5.2's ensemble at past cutoffs	<code>history (resumable; 215 of 955 origins done)</code>	no	CPU, hours
Fit omega on that history	<code>fit_width</code>	no	free, minutes
Weekly review, zero spend	<code>review.run_weekly --no-llm</code>	no	free
Weekly review, real	<code>review.run_weekly [--dry-run --bootstrap]</code>	yes	\$1 to 5, \$10 bootstrap
The coached forecast	<code>review.coached [--stream ...] (weekdays 12:00 once its plist is installed)</code>	no	free
Apply a decision	<code>review.accept --proposal P-0003 --stream v52_lite --by <you></code>	no	free
Champion vs challenger	<code>review.pairing --champion v52_advanced --challenger v53_<slug>_advanced</code>	no	free

Reading the report

- **Fidelity line first.** "all N records replay exactly" means every score below is trustworthy. FIDELITY DRIFT means stop and explain it.
- **Read the live_forward band, not the top line.** Replayed and backfilled records are scored and shown but excluded from every fit.
- **Four variants, one set of days.** `random_walk` is the floor, `ensemble` is the models alone, `agent` is what was published, `coached` (or `current`) is the same runs under the coach ledger. Because they share days, a difference is a difference in calibration.
- **Effective n, not n.** Overlapping horizons mean 18 daily cutoffs are about 3.4 independent observations. The report prints that number beside every verdict and labels underpowered passes.

Deciding on a proposal

Read the weekly report.md top down. The scope line says what can and cannot be concluded this week. The decomposition says where the loss sits. Each proposal names its track, its effect against the random walk per horizon, and the file that would be applied. To decide, edit the decision block of the proposal's YAML: status (accepted, rejected, deferred), by, on, reason. Then run `accept.py`. Numeric acceptances change the coached forecast from the next business day. Text acceptances produce a challenger directory with a REGISTER.md to apply by hand. Code briefs are implemented in an engineering session.

Two operational rules

Each `live_forward` record is irreplaceable. A missed weekday cannot be recovered, because re-running it tomorrow yields a replayed record. And the attempt watchdog counts awake time only, so a laptop that sleeps through the 08:00 job produces late or missing records. Keep one machine awake and on power, and let it own the run directories.

7 | WHERE IT STANDS

The scoreboard today, what is built, and what is missing

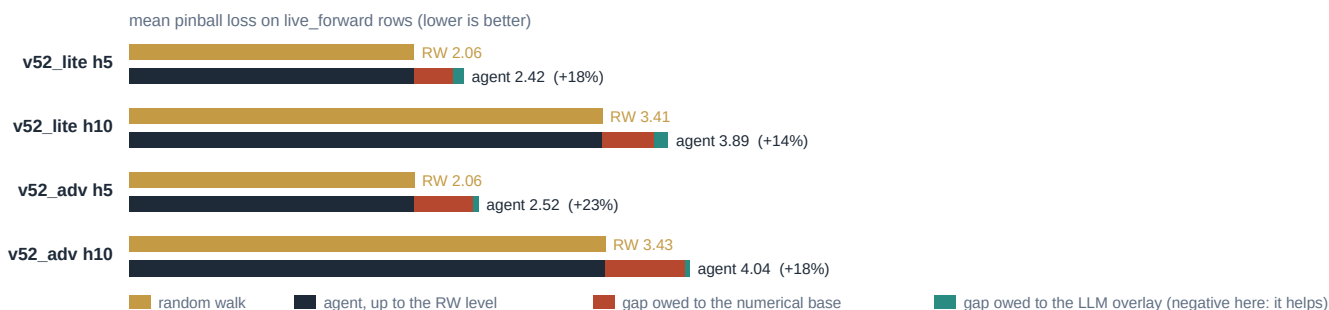


Figure 6. Live_forward rows with prices through 2026-09-11: 18 cutoffs per stream, one rally episode, effective n about 3.4. The red segment is the part of the gap the ensemble alone would have shown; the overlay's share is small and negative. Coverage is 31 to 38% against an 80% target; the random walk covers 45 to 63%.

Area	Built and verified	Designed, not yet done
Recording and replay	Three streams, 288 records, exact fidelity, schema versioning, backfill of pre-coach audit files	Numerical fidelity: replay starts from stored ensemble quantiles, so a non-deterministic LightGBM or a silently dropped model is invisible to it
Scoring	Pinball, CRPS, 80% coverage, direction vs always-up, random-walk baseline, loss decomposition, effective n	Direction significance is unreachable at this sample; the report says so rather than printing a hit rate as skill
Calibration levers	Layer with three per-horizon fields, settings overlay with identity denylist, coach-owned ledger, coached corpus (232 records, identity ledger)	centre_gain fit blocked until same-cutoff action variance is measured (window runs to 2026-10-30)
The gate	Ten conditions, origin-blocked bootstrap, shrinkage, holdout by fitted_through	No candidate has passed: 18 origins exist but the two statistical conditions are underpowered below effective n of 10
History and omega	history.py re-runs v5.2's suite at weekly origins since 2008 with cutoff-honest data; fit_width fits omega per horizon	History is 215 of 955 origins complete; fit_width has not run on the full range; the ensemble_only provenance decision for width fits is argued but not taken in config
Weekly review	All eight stages, taxonomy, seeds, memory, gates, challenger drafting, pairing, accept, ship watch, self-metrics; first live run 2026-09-13 at \$0.89	Bootstrap run waits on history and fit_width; the weekly and coached-forecast plists are written but not installed
Trust report	Tier logic and a 4,700-day market case base exist in the tree	Not wired in: it returned high for runs that missed by \$27 to 35. It stays out until it can be ranked against realized error

What is missing

- The biggest loss is not the coach's to fix.** LightGBM predicts price levels it cannot extrapolate and sits below the last close on 21 of 21 cutoffs. The numeric levers can mask this (rw_anchor, weights); only a code change to the agent removes it. That brief exists and is unimplemented.
- The overlay signal is noisy in ways that block a centre fit.** The same cutoff run twice can publish +\$3.58 and \$0.00 because citation wiring failed the policy. The coach can detect this (a promoted process hypothesis) but cannot change the skill text except through a challenger that nobody has registered yet.
- Evidence accrues slowly.** Outcome-dependent LLM proposals need roughly 8 to 10 more weeks of cutoffs; a challenger verdict needs 12 shared cutoffs plus the 21-day lag. Nothing in the design shortens that.
- Publishing is manual.** The corpus page and the weekly HTML are regenerated and shared from a session, not served.
- The coach judges with the model it judges.** The review LLM is the same family as the advanced stream. Code-computed counts, pointer validation, blind re-triage and cross-stream contrast mitigate this; they do not remove it.

8 | GLOSSARY

Terms as the code and the reports use them

Stream	One (agent package, LLM, runs per day) with its own run directory and calibration ledger. v52_lite (10 runs a day on the lite model), v52_advanced (3 a day on gemini-3.5-flash), v50_lite (retired 2026-09-08, still scored).	Pinball loss	Mean quantile loss over the quantile grid; the primary metric. Additive, so a gap decomposes into a base term and an overlay term.
Run record	The stored snapshot of one agent execution, complete enough to replay under any calibration.	CRPS	Continuous ranked probability score; recorded so coach numbers stay comparable with the repo leaderboard.
Cutoff / origin	The date a forecast is issued from; the last close visible to it is the previous business day. Sample sizes are counted in origins.	Coverage (cover80)	Fraction of outcomes inside P10..P90. Target 80%.
Horizon (h)	Business days ahead: 5, 10 and 21. A run has one distribution per horizon.	Candidate	A proposed calibration awaiting the gate, with fitted_through declaring where its holdout begins.
Provenance	How a record's research was obtained. live_forward (today's cutoff, evidence), replayed_live_search (a past cutoff, not evidence), backfill_cached_news, ensemble_only (no LLM), coached_replay (coach output, never evidence).	Incumbent	The version in force that a candidate is compared against.
Fidelity	Replaying a record under its own calibration reproduces its published quantiles exactly. Checked before any scoring.	Holdout	Origins after fitted_through. A candidate fitted on everything has none and cannot pass.
Calibration version	A dated JSON document with a settings overlay and a layer. Immutable once written. v001 is the empty baseline in every ledger.	Shrinkage	The passed value is moved halfway from the incumbent toward the candidate before a human sees it.
Calibration ledger	The directory of versions for one stream, with a lookup of which version was in force on a date. Live ledgers (calibration_v52_*) stay at v001; the coach writes only its own (review/calibration/<stream>/).	Effective n	Independent observations after accounting for overlapping horizons. Verdicts below 10 are labelled underpowered.
Settings overlay	The v5.2 settings fields a version overrides. May never name a model, timeout, or audit flag.	Episode	A same-direction move in the price path, cut by a zig-zag whose threshold is the random walk's h10 spread.
Layer	The coach's post-engine transform: centre_gain, width_scale (omega) and rw_anchor, each per horizon.	Independent window	Supporting cutoffs at least h business days apart, so their outcome windows do not overlap.
Overlay (LLM overlay)	The dollar distance the agent moved the centre away from the raw ensemble, after the evidence policy granted or capped the LLM's proposed action.	Case card	The code-built, cached view of one (stream, cutoff) that triage reads: representative run, dispersion, numerical base.
Evidence policy / tier	v5.2's code that turns a proposed action plus its citations into a granted action. Tiers none, limited, corroborated, strong depend on resolved publishers; tier none zeroes the overlay.	Track	Which proposal path a tag feeds: numeric, policy, llm, data, or none (a no-fault regime shock).
Ensemble	ARIMA + Kalman + LightGBM quantiles with no LLM. The floor the agent must beat, and what a neutral run publishes unchanged.	Hypothesis / seed	A pattern with support counted by code. Seeds come from IMPROVEMENTS.md and are hidden from triage.
Random walk (RW)	The baseline: centre on the last close, width from recent spread. The target the agent must beat on pinball.	Proposal	P-NNNN.yaml: statement, mechanism, lever, evidence, effect vs RW, and the decision block a person edits.
		Challenger	A drafted copy of v5.2 with a text edit applied, its own manifest and stream, compared with the champion by pairing on shared cutoffs.
		Coached forecast	Each day's v5.2 records replayed under the coach ledger and written as separate records; the report's coached variant.
		Package fingerprint	SHA-256 over the agent's manifest. A changed build is a different agent and cannot be pooled with the old one.
		Prompt version	Identifier for the prompt builder in force. Records across prompt versions are never pooled.